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The effects of vitamin D supplementation on wound healing and metabolic status in patients with diabetic foot ulcer: A randomized, double-blind, placebo-controlled trial

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ABSTRACT

Objective: This study was conducted to evaluate the effects of vitamin D supplementation on wound healing and metabolic status in patients with diabetic foot ulcer (DFU).
Methods: This randomized, double-blind, placebo-controlled trial was performed among 60 patients with grade 3 DFU according to "Wagner–Meggitt's" criteria. Participants were randomly divided into two groups (each 30 participants) and received either 50,000 IU vitamin D supplements every 2 weeks for 12 weeks (group A) or placebo (group B). Fasting blood samples were taken at study baseline and after 12-week intervention to determine related markers.
Results: After 12 weeks of intervention, compared with the placebo, vitamin D supplementation resulted in a significant reduction in ulcer length (−2.1 ± 1.1 vs. −1.1 ± 1.1 cm, P = 0.001), width (−2.0 ± 1.2 vs. −1.1 ± 1.0 cm, P = 0.02) and depth (−1.0 ± 0.5 vs. −0.5 ± 0.5 cm, P < 0.001), and erythema rate (100% vs. 80%, P = 0.01). In addition, in supplemented patients changes in serum insulin concentration (−3.4 ± 9.2 vs. +2.8 ± 9.3 μIU/mL, P = 0.01), homeostasis model of assessment-estimated insulin resistance (−1.5 ±4.1 vs. +1.7 ± 5.1, P = 0.01), the quantitative insulin sensitivity check index (+0.006 ± 0.02 vs. −0.006 ±0.02, P = 0.03) and HbA1c (−0.6 ± 0.6 vs. −0.1 ± 0.5%, P = 0.004) were significantly different from those of patients in the placebo group. Additionally, following supplementation with vitamin D, significant reductions in serum total- (−15.8 ± 18.9 vs. +5.3 ± 31.8 mg/dL, P = 0.003), LDL- (−17.2 ± 19.8 vs. +2.2 ± 28.6 mg/dL, P = 0.003), total-/HDL-cholesterol ratio (−1.1 ± 0.8 vs. −0.2 ± 1.1, P = 0.001), high sensitivity C-reactive protein (hs-CRP) (−0.4 ± 2.5 vs. +1.9 ± 4.2 μg/mL, P = 0.01), erythrocyte sedimentation rate (ESR) (−34.7 ± 32.4 vs. −18.0 ± 26.6 mm/h, P = 0.03) and plasma malondialdehyde (MDA) concentrations (−0.7 ± 0.9 vs. −0.2 ± 0.5 μmol/L, P = 0.008) were seen compared with the placebo.
Conclusions: Overall, vitamin D supplementation for 12 weeks among patients with DFU had beneficial effects on glucose homeostasis, total-, LDL-, total-/HDL-cholesterol, ESR, hs-CRP and MDA levels. In addition, vitamin D may have played an indirect role in wound healing due to its effect on improved glycemic control.

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1. Introduction

The diabetic foot ulcer (DFU) is a common complication of diabetes mellitus affecting 10%–25% of people with diabetes (Armstrong et al., 2014). The annual incidence of ulceration and amputation among patients with DFU is as high as 2.5%–10.7% and 0.25%–1.8%, respectively (Tiaka, Papanas, Manolakis, & Maltezos, 2011). Patients with diabetic foot ulcer have a higher mortality compared with patients with diabetes who have not developed food ulcer. DFU is a major cause of morbidity in diabetic patients, and the mortality rate is about twice that of diabetic patients without foot ulcer (Daousi et al., 2004). Diabetic neuropathy and peripheral vascular disease are the main etiological factors in DFU (Sinwar, 2015). Furthermore, insulin resistance, dyslipidemia, inflammation and oxidative stress play a significant role in the pathogenesis of DFU (Karadurmus et al., 2010; Sytze Van Dam, Cotter, Bravenboer, & Cameron, 2013).

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In recent years, there has been an effort to understand possible roles of vitamin D inadequacy including its role in pancreatic insulin secretion and insulin action, the immune system particularly on T cell medicated immunity and decreased inflammation and oxidative stress (Asemi, Hashemi, Karamali, Samimi, & Esmaillzadeh, 2013; Zubair, Malik, Meerza, & Ahmad, 2013). Circulating levels of 25(OH)D were low in patients with DFU (Asemi et al., 2013; Tiwari et al., 2013) and the beneficial effects of vitamin D supplementation on markers of insulin resistance, biomarkers of inflammation and oxidative among patients without DFU were reported. We have previously indicated that supplementation with 50,000 IU vitamin D/week among patients with major depressive disorder for 8 weeks had the beneficial effects on parameters of glucose homeostasis, and oxidative stress, but did not affect lipid profiles (Sepehrmanesh et al., 2016). However, improvement in vitamin D status following administration with 280 μg/day for 2 weeks and 140 μg/day for 10 weeks did not improve insulin resistance among patients with diabetes (Kampmann et al., 2014).

The favorable effects of vitamin D supplementations on wound healing, insulin resistance, biomarkers of inflammation and oxidative stress may be mediated by its impact on stimulating the phagocytosis and killing the bacteria by macrophages (van Etten, Decallonne, Bouillon, & Mathieu, 2004), suppressing interferon-g-mediated macrophage activation (Helming et al., 2005), activating insulin receptor expression, and the downregulation of cytokine generation (Pittas, Lau, Hu, & Dawson-Hughes, 2007). As there is evidence that vitamin D intake may accelerate wound healing and has anti-inflammatory and antioxidant effects, we hypothesized that vitamin D supplementation might help patients with DFU to heal their wound faster, and have better metabolic profiles, and biomarkers of inflammation and oxidative stress. The objective of this study was to evaluate the effects of vitamin D supplementation on wound healing and metabolic status in patients with DFU.

2. Methods

2.1. Trial design

The current study was a prospective randomized double-blind placebo-controlled clinical trial.

2.2. Participants

In the current study, 60 patients with grade 3 DFU aged 40–85 years who referred to the Shahid Beheshti Clinic in Kashan, Iran, between November 2015 and January 2016 were recruited. The Wagner–Meggitt's original system has six grades of lesions. This system assesses ulcer depth and the presence of osteomyelitis or gangrene by using the following grades: grade 0 (pre-or postulcerative lesion), grade 1 (partial/full thickness ulcer), grade 2 (probing to tendon or capsule), grade 3 (deep ulcer with abscess or osteomyelitis), grade 4 (partial foot gangrene), and grade 5 (whole foot gangrene) (Jain, 2012). In the current study, all patients had the same grade 3. Pregnant and breastfeeding patients, participants who consumed vitamin D supplements during the past 3 months, anticipated changes in medications throughout the study and patients with history of diseases which influence the development of DFU including chronic trauma were excluded.

2.3. Ethics statements

This trial was conducted in accordance with the Declaration of Helsinki and informed consent was received from all patients. The research was approved by the ethics committee of Kashan University of Medical Sciences (KUMS) and was recorded in the Iranian website for registration of clinical trials (http://www.irct.ir: IRCT201510315623N54).

2.4. Study design

At the onset of the study, all participants were stratified for gender (males: 22 and 8 females in each group), type and dosage of medications, duration of diabetes mellitus (DM), pre-intervention BMI (<25 and ≥25 kg/m^2) and age (<55 and ≥55 years). Participants were then randomly divided into two groups to receive either vitamin D supplements (n = 30) or placebo (n = 30) for 12 weeks.

In addition, all participants underwent a similar treatment protocol for the Diabetic Foot, based on the Infectious Diseases Society of America (IDSA) (Berbari et al., 2015). Participants were requested not to change their ordinary physical activity and not to take any nutritional supplements during the 12-week trial. All patients completed 3-day food records and three physical activity records at the study baseline, weeks 3, 6 and 9 of the intervention and end-of-the trial. Daily macro- and micro-nutrient intakes were analyzed by nutritionist IV software (First Databank, San Bruno, CA). In the current study, physical activity was described as metabolic equivalents (METs) in hours per day. To determine the METs for each patient, we multiplied the times (in hour per day) reported for each physical activity by its related METs coefficient by standard tables (Ainsworth et al., 2000).

2.5. Intervention

In the intervention group, patients received 50,000 IU vitamin D supplements every 2 weeks for 12 weeks. Vitamin D supplements and placebos capsules were similar in shape and size and manufactured by Zahravi (Tabriz, Iran).

2.6. Treatment adherence

Every four weeks, participants were given enough supplements to last until 3 days after their next scheduled visit and were instructed to return all the unused supplements at each visit. Patients were scheduled for the follow-up visits every 2 weeks at weeks 2, 4, 6, 8, 10 and 12 of the intervention for an intermediate evaluation and an ulcer debridement as well as to complete 3-day food records and three physical activity records at weeks 0, 3, 6, 9 and 12 of the intervention. To determine the compliance the remaining supplements were counted and subtracted from the amount of supplements provided to the participants. To increase compliance, all participants received short messages on their cell phones every day to remind them about taking the capsules.

2.7. Assessment of anthropometric measures

Weight and height of participants were determined in an overnight fasting status using a standard scale (Seca, Hamburg, Germany) at the onset of the study and after 12-weeks' intervention. BMI was calculated as weight in kg divided by height in meters squared.

2.8. Assessment of outcomes

In our study, wound healing and markers of insulin resistance were considered as the primary outcome and lipid profiles, and biomarkers of inflammation and oxidative stress were considered as the secondary outcomes.

2.9. Clinical assessment

Ulcer size was measured by multiplying the largest by the second largest perpendicular diameter of the skin lesion. Cumulative ulcer size (sum of the two largest perpendicular diameters and ulcer depth) was also computed for each patient. Depth of the ulcer was defined as

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either superficial or deep. Infection was diagnosed if edema, erythema, discharge, regional lymph node enlargement, pain, or fever was present (Berbari et al., 2015). Patients were considered positive for infection if at least two of the above listed features were present. According to their location, ulcers were defined as plantar or nonplantar. Each foot was assessed and graded according to the Wagner classification (Berbari et al., 2015).

2.10. Biochemical assessment

Ten milliliters of fasting blood samples was collected at the beginning and the end of the study at Kashan reference laboratory in a fasting status and centrifuged to separate serum. Then, the samples were stored at −80 °C before analysis. Serum 25-hydroxyvitamin D concentrations were quantified by a commercial ELISA kit (IDS, Boldon, UK) with inter- and intra-assay coefficient variances (CVs) of 4.3% to 6.5%, respectively. Serum insulin concentrations were assessed using available ELISA kit (DiaMetra, Milano, Italy). The homeostasis model of assessment-insulin resistance (HOMA-IR), β-cell function (HOMA-B) and the quantitative insulin sensitivity check index (QUICKI) were determined according to the suggested formulas (Pisprasert, Ingram, Lopez-Davila, Munoz, & Garvey, 2013). Hemoglobin A1c ($HbA1_C$) levels in the whole blood were quantified by Glycomat kit (BiocodeHycel, Massy, France) using the method of exchange chromatography. Enzymatic kits (Pars Azmun, Tehran, Iran) were used to quantify FPG, serum triglycerides, VLDL-, total-, LDL- and HDL-cholesterol concentrations. All inter- and intra-assay CVs for FPG and lipid concentrations were less than 5%. Erythrocyte sedimentation rate (ESR by Westergren method) was obtained at baseline and 12 weeks after starting treatment. Serum high sensitivity C-reactive protein (hs-CRP) concentrations were evaluated by commercial ELISA kit (LDN, Nordhorn, Germany). The plasma nitric oxide (NO) concentrations were assessed using Griess method (Tatsch et al., 2011). Plasma total antioxidant capacity (TAC) concentrations were determined by the method of ferric reducing antioxidant power developed byBenzie and Strain (1996), total glutathione (GSH) was determined using the method of Beutler and Gelbart (1985) and malondialdehyde (MDA) concentrations were determined by the thiobarbituric acid reactive substances spectrophotometric test (Janero, 1990). All inter- and intra-assay CVs for NO, TAC, GSH and MDA concentrations were less than 5%.

2.11. Sample size

Using a formula suggested for clinical trials, having 25 patients in each group was adequate while considering a type one error (α) of 0.05 and type two error (β) of 0.20 (power = 80%), 1.28 as SD and 1.00 as the mean distinction (d) of HOMA-IR as the key variable (Asemi et al., 2013). Assuming 5 dropouts in each group, the final sample size was determined to be 30 patients in each group.

2.12. Randomization

Randomization assignment was performed using computer-generated random numbers. Randomization and allocation were concealed from the researchers and participants until the final analyses were completed. The randomized allocation sequence, enrolling participants and allocating them to interventions were conducted by a trained staff at the clinic.

2.13. Statistical methods

To evaluate whether the study variables were normally distributed or not, we used the Kolmogrov–Smirnov test. Analyses were conducted based on an intention-to-treat (ITT) principle. Missing values were treated based on Last-Observation-Carried-Forward method (LOCF) (Lachin, 2016). LOCF ignores whether the participant's condition was improving or deteriorating at the time of dropout but instead freezes outcomes at the value observed before dropout (i.e., last observation) (Lachin, 2016). To detect differences in anthropometric measures as well as in macro- and micro-nutrient intakes between the two groups, we applied Student's t test to independent samples. To determine the effects of vitamin D administration on ulcer size, markers of insulin resistance, lipid profiles, biomarkers of inflammation and oxidative stress, we used one-way repeated measures analysis of variance. In this analysis, the treatment (vitamin D vs. placebo group) was regarded as a between-subject factor and time with two time-points (the onset of the study and after 12-intervention) was considered as within-subject factor. To identify within-group differences (baseline and end-of-trial), we used paired-samples t-tests. Differences in proportions were evaluated by Chi square test or Fisher's exact tests. In addition, to identify within-group differences (baseline and end-of-trial) of nominal variables, we used the non-parametric McNemar test. Adjustment for changes in baseline values of biochemical parameters, age and BMI at the baseline was performed by analysis of covariance (ANCOVA) using general linear models. The P-value of <0.05 was considered statistically significant. All statistical analyses used the Statistical Package for Social Science version 18 (SPSS Inc., Chicago, IL, USA).

3. Results

Among participants in the placebo group, 2 patients [withdrawn for personal reasons (n = 2)] were excluded (Fig. 1). At the end, 58 patients with DUF [vitamin D (n = 30) and placebo (n = 28)] completed the trial. However, as the analysis was based on the ITT principle, all 60 patients (30 in each group) were included in the final analyses.

Gender distribution, mean age, weight and BMI at the baseline and end-of-trial, METs at the baseline and end-of-trial, and insulin and metformin therapy were not statistically different between the two groups (Table 1). Mean height of study participants in the vitamin D group was taller than in the placebo group (173.6 ± 7.3 vs.169.9 ± 6.2 cm, P = 0.03).

Based on the 3-day dietary records obtained at the baseline, end-of-trial and throughout the trial, we found no significant difference in mean dietary macro- and micro-nutrient intakes between the two groups (Table 2).

After 12 weeks of intervention, compared with the placebo, vitamin D supplementation resulted in a significant reduction in ulcer length (−2.1 ± 1.1 vs. −1.1 ± 1.1 cm, P = 0.001), width (−2.0 ± 1.2 vs. −1.1 ± 1.0 cm, P = 0.02) and depth (−1.0 ± 0.5 vs. −0.5 ± 0.5 cm, P < 0.001) (Fig. 2). In addition, vitamin D supplementation led to a significant decrease in HOMA-IR (−1.5 ± 4.1 vs. +1.7 ± 5.1, P = 0.01) compared with the placebo.

Compared with the placebo, vitamin D supplementation led to a significant rise in serum 25-hydroxyvitamin D levels (+12.9 ± 10.0 vs. −1.8 ± 15.7 ng/mL, P < 0.001) (Table 3). In addition, in supplemented patients changes in serum insulin concentration (−3.4 ± 9.2 vs. +2.8 ± 9.3 μIU/mL, P = 0.01), HOMA-B (−7.7 ± 23.7 vs. +4.2 ± 18.8, P = 0.03), QUICKI (+0.006 ± 0.02 vs. −0.006 ± 0.02, P = 0.03) and HbA1c (−0.6 ± 0.6 vs. −0.1 ± 0.5%, P =0.004) were significantly different from those of patients in the placebo group. Additionally, following supplementation with vitamin D, significant reductions in serum total- (−15.8 ± 18.9 vs. +5.3 ± 31.8 mg/dL, P = 0.003), LDL- (−17.2 ± 19.8 vs. +2.2 ± 28.6 mg/dL, P = 0.003), total-/HDL-cholesterol ratio (−1.1 ± 0.8 vs. −0.2 ± 1.1, P = 0.001), hs-CRP (−0.4 ± 2.5 vs. +1.9 ± 4.2 μg/mL, P = 0.01), ESR (−34.7 ± 32.4 vs. −18.0 ± 26.6 mm/h, P = 0.03) and plasma MDA concentrations (−0.7 ± 0.9 vs. −0.2 ± 0.5 μmol/L, P = 0.008) were seen compared with the placebo. There were no significant

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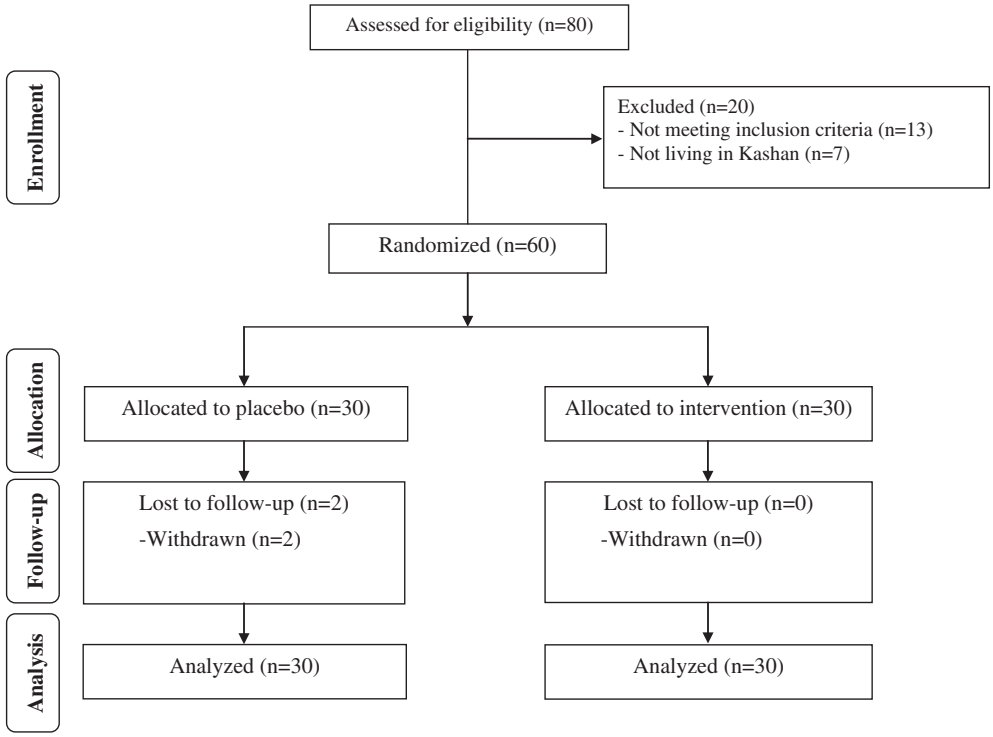

Fig. 1. Summary of patient flow diagram.

changes in FPG, other lipid profiles, biomarkers of inflammation and oxidative stress between the two groups.

When we adjusted the analysis for baseline values of biochemical parameters, age and baseline BMI, no significant change in our findings was observed except for FPG (P = 0.01) and HOMA-B (P = 0.09) (Table 4). Further adjustment for insulin and metformin therapy did not influence our findings.

4. Discussion

In this trial, which to the best of our knowledge is the first of its kind, we evaluated effects of vitamin D supplementation on wound healing, glycemia status, lipid concentrations, and biomarkers of inflammation and oxidative stress among patients with DFU. We demonstrated that vitamin D supplementation for 12 weeks among patients with DFU had beneficial effects on glucose homeostasis, total-, LDL-, total-/HDL-cholesterol, ESR, hs-CRP and MDA levels, but it did not had any effect on FPG, other lipid profiles, and biomarkers of inflammation and oxidative stress. In addition, vitamin D may have played an indirect role in wound healing due to its effect on improved glycemic control.

We found that vitamin D supplementation among patients with DFU for 12 weeks resulted in a significant improve in parameters of wound healing compared with the placebo. Few observational studies have reported that the prevalence and severity of vitamin D deficiency were high in patients with DFU (Tiwari, Pratyush, Gupta, & Singh, 2014; Tiwari et al., 2013). Hyperglycemia in diabetic patients disrupts the normal process of cytokine production (Naguib, Al-Mashat, Desta, & Graves, 2004), which in turn leads to the development of chronic wound (Agren et al., 2000). In addition, several physiological factors such as decreased growth factor production (Galkowska, Wojewodzka, & Olszewski, 2006), macrophage function (Maruyama et al., 2007) and

Table 1
General characteristics of study participants.

	Placebo group (n = 30)	Vitamin D group (n = 30)	P Value[a]
Gender (%)			
Male	22 (73.3)	22 (73.3)	1.00[b]
Female	8 (26.7)	8 (26.7)	
Age (y)	58.6 ± 8.6	59.6 ± 8.2	0.64
Height (cm)	169.9 ± 6.2	173.6 ± 7.3	0.03
Weight at study baseline (kg)	75.5 ± 11.8	78.5 ± 14.3	0.37
Weight at end-of-trial (kg)	75.6 ± 11.8	78.6 ± 14.2	0.39
Weight change (kg)	0.1 ± 1.8	0.1 ± 1.1	0.76
BMI at study baseline (kg/m^2)	26.2 ± 3.8	26.0 ± 4.4	0.90
BMI at end-of-trial (kg/m^2)	26.2 ± 3.9	26.0 ± 4.3	0.86
BMI change (kg/m^2)	0.05 ± 0.6	0.008 ± 0.4	0.70
MET-h/day at study baseline	26.0 ± 2.5	26.8 ± 2.7	0.27
MET-h/day at end-of-trial	25.8 ± 2.6	26.8 ± 2.8	0.19
MET-h/day change	−0.1 ± 0.8	−0.01 ± 0.9	0.44
Insulin therapy (%)	30 (100.0)	30 (100.0)	1.00[b]
Metformin therapy (%)	27 (90.0)	26 (86.7)	0.68[b]

Data are means ± SDs.
METs, metabolic equivalents.
[a] Obtained from independent t test.
[b] Obtained from Pearson Chi-square test.

Table 2
Dietary intakes of study participants throughout the study.

	Placebo group (n = 30)	Vitamin D group (n = 30)	P Value[a]
Energy (kcal/d)	2165 ± 210	2238 ± 217	0.19
Carbohydrates (g/d)	299.6 ± 43.0	310.3 ± 50.0	0.37
Protein (g/d)	80.3 ± 19.1	80.9 ± 11.2	0.89
Fat (g/d)	75.7 ± 13.8	78.3 ± 14.0	0.46
SFAs (g/d)	23.5 ± 4.9	24.6 ± 4.6	0.35
PUFAs (g/d)	23.3 ± 5.5	24.3 ± 6.3	0.54
MUFAs (g/d)	20.7 ± 6.0	21.3 ± 5.6	0.70
Cholesterol (mg/d)	188.0 ± 106.3	198.0 ± 106.3	0.71
TDF (g/d)	19.7 ± 4.3	18.3 ± 4.3	0.22
Vitamin D (μg/d)	2.9 ± 0.8	3.0 ± 0.9	0.69
Calcium (mg/d)	1073.4 ± 146.9	1116.6 ± 185.7	0.32

Data are means ± SDs.
MUFAs, monounsaturated fatty acids; PUFAs, polyunsaturated fatty acids; SFAs, saturated fatty acids; TDF, total dietary fiber.
[a] Obtained from independent t test.

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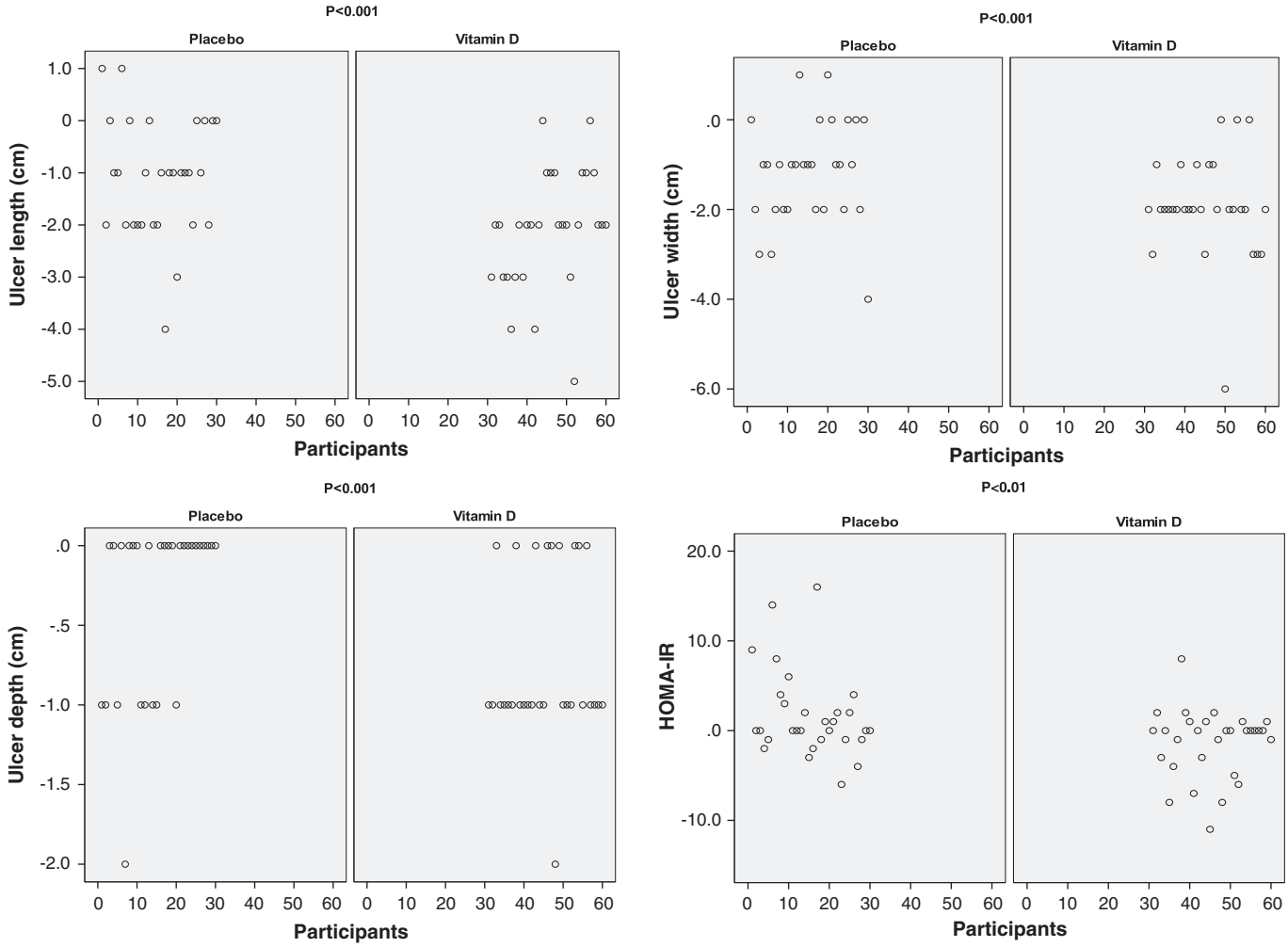

Fig. 2. Scatter-gram of mean changes in ulcer size and HOMA-IR after 12 weeks of intervention. HOMA-IR, homeostasis model of assessment-estimated insulin resistance.

impaired cytokine production (Galiano et al., 2004) contribute to wound-healing abnormalities in individuals with diabetes. Previous studies have suggested that vitamin D deficiency is associated with increased susceptibility towards infection (Tiwari et al., 2013). Furthermore, it has been shown that active vitamin D3 stimulates the phagocytosis and killing of the intestine bacteria by macrophages (van Etten et al., 2004) and is a potent suppressor of the interferon-g-mediated macrophage activation (Lemire., 1992). Vitamin D also suppresses T cell proliferation and decreases the production of T helper type 1 cytokines, while promoting the production of T helper type 2 cytokines (van Etten & Mathieu, 2005), which may improve parameters of wound healing.

Findings of the current study demonstrated that vitamin D administration compared with the placebo in patients with DFU for 12 weeks led to significant reductions in serum insulin, HOMA-IR, HOMA-B, HbA1c, serum total- and LDL- and total-/HDL-cholesterol, and a significant increase in QUICKI, but did not affect FPG and other lipid profiles. Observational studies have indicated that low 25-hydroxyvitamin D [25(OH)D] levels were associated with impaired in glucose homeostasis parameters (Ferreira Tda, Rocha, Klein, & Sanjuliani, 2015). It is difficult to determine a solid rationale for such associations, unless there was an effect of vitamin D administration on markers of insulin resistance in placebo-controlled randomized trials. Unfortunately, the intervention studies gave controversial results with some showing a positive and some a negative effect. We have previously shown that supplementation with 50,000 IU vitamin D/every two weeks for 12 weeks among women at risk of pre-eclampsia improved markers of insulin resistance (Karamali, Beihaghi, Mohammadi, & Asemi, 2015). However, supplementation with 1000 IU/day vitamin D among healthy overweight or obese women for 12 weeks did not influence insulin resistance (Salehpour et al., 2013). About the effects of vitamin D on lipid profiles, several experimental and clinical studies previously demonstrated that vitamin D supplements have decreasing effects on serum lipid profiles. Supplementation with 4000 IU/day vitamin D in postmenopausal women with type 2 diabetes mellitus (T2DM) for 6 months resulted in a significant decrease in serum triglyceride levels without affecting levels of other lipid profiles (Munoz-Aguirre et al., 2015). No significant change in circulating levels of triglycerides, LDL- and HDL-cholesterol was observed in patients with T2DM following supplementation with 16,000 IU of vitamin D orally once a week for 8 weeks (Ramiro-Lozano & Calvo-Romero, 2015). Favorable effects of vitamin D intake on markers of insulin resistance might be explained by its effect on calcium and phosphorus metabolism and through up-regulation of the insulin receptor genes (Maestro, Molero, Bajo, Davila, & Calle, 2002) and increased transcription of insulin receptor genes (Maestro et al., 2002). Moreover, few mechanisms could explain the effect of vitamin D on decreasing concentrations of lipid and lipoproteins. Vitamin D intake might increase calcium absorption, which in turn would result in increasing fat absorption (Reid, 2004). In addition, vitamin D through improved insulin sensitivity might mediate its potential benefit on lipid profiles (Rajpathak et al., 2010).

This study showed that compared with the placebo, taking vitamin D supplements for 12 weeks among patients with DFU decreased serum hs-CRP, ESR and plasma MDA concentrations, but did not

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Table 3
Wound healing, metabolic profiles, and biomarkers of inflammation and oxidative stress at the baseline and after 12 weeks of the intervention in patients with diabetic foot ulcer.

	Placebo group (n = 30)				Vitamin D group (n = 30)				P Value[b]
	Baseline	End-of-trial	Change	P Value[a]	Baseline	End-of-trial	Change	P Value[a]	
Vitamin D (ng/mL)	20.2 ± 15.6	18.4 ± 8.5	−1.8 ± 15.7	0.52	15.2 ± 9.9	28.1 ± 12.5	12.9 ± 10.0	<0.001	<0.001
Ulcer symptoms									
Erythema (%)	30 (100)	6 (20)	–	<0.001[c]	30 (100)	0 (0)	–	<0.001[c]	0.01[d]
Discharge (%)	30 (100)	5 (16.7)	–	<0.001[c]	30 (100)	1 (3.3)	–	<0.001[c]	0.08[d]
Necrosis (%)	9 (30)	3 (10)	–	0.002[c]	6 (20)	0 (0)	–	0.03[c]	0.18[d]
FPG (mg/dL)	187.8 ± 70.7	174.4 ± 68.5	−13.4 ± 48.3	0.13	171.8 ± 62.2	140.5 ± 31.9	−31.3 ± 46.4	0.001	0.14
Insulin (μIU/mL)	15.0 ± 6.3	17.8 ± 8.5	2.8 ± 9.3	0.10	17.1 ± 9.4	13.7 ± 6.4	−3.4 ± 9.2	0.05	0.01
HOMA-B	29.4 ± 20.1	33.6 ± 19.7	4.2 ± 18.8	0.23	36.9 ± 29.0	29.2 ± 20.1	−7.7 ± 23.7	0.08	0.03
QUICKI	0.29 ± 0.01	0.29 ± 0.02	−0.006 ± 0.02	0.18	0.30 ± 0.03	0.30 ± 0.002	0.006 ± 0.02	0.15	0.03
HbA1c (%)	7.8 ± 0.7	7.7 ± 0.6	−0.1 ± 0.5	0.20	8.3 ± 1.0	7.7 ± 0.5	−0.6 ± 0.6	<0.001	0.004
Triglycerides (mg/dL)	154.2 ± 93.1	151.7 ± 104.0	−2.5 ± 36.7	0.71	163.4 ± 66.5	147.5 ± 50.7	−15.9 ± 25.1	0.002	0.10
VLDL-cholesterol (mg/dL)	30.8 ± 18.6	30.3 ± 20.8	−0.5 ± 7.3	0.71	32.7 ± 13.3	29.5 ± 10.1	−3.2 ± 5.0	0.002	0.10
Total cholesterol (mg/dL)	159.4 ± 45.0	164.7 ± 49.2	5.3 ± 31.8	0.37	177.5 ± 35.3	161.7 ± 31.2	−15.8 ± 18.9	<0.001	0.003
LDL-cholesterol (mg/dL)	96.5 ± 38.1	98.7 ± 41.3	2.2 ± 28.6	0.68	112.1 ± 31.5	94.9 ± 30.3	−17.2 ± 19.8	<0.001	0.003
HDL-cholesterol (mg/dL)	32.0 ± 4.5	35.6 ± 6.2	3.6 ± 6.5	0.005	32.7 ± 3.6	37.4 ± 3.9	4.6 ± 4.1	<0.001	0.46
Total-/HDL-cholesterol ratio	5.0 ± 1.3	4.8 ± 1.8	−0.2 ± 1.1	0.27	5.5 ± 1.2	4.4 ± 1.0	−1.1 ± 0.8	<0.001	0.001
hs-CRP (μg/mL)	15.4 ± 5.6	17.3 ± 5.2	1.9 ± 4.2	0.01	13.2 ± 3.9	12.8 ± 4.2	−0.4 ± 2.5	0.34	0.01
ESR (mm/h)	64.5 ± 36.8	46.5 ± 30.7	−18.0 ± 26.6	0.001	67.1 ± 35.8	32.4 ± 19.7	−34.7 ± 32.4	<0.001	0.03
NO (μmol/L)	43.1 ± 4.6	44.8 ± 5.7	1.7 ± 6.8	0.17	40.6 ± 8.9	45.0 ± 5.3	4.4 ± 9.6	0.01	0.22
TAC (mmol/L)	1295.6 ± 134.2	1200.8 ± 180.5	−94.8 ± 172.6	0.005	1314.1 ± 98.8	1219.2 ± 180.5	−94.9 ± 192.6	0.01	0.99
GSH (μmol/L)	540.0 ± 98.3	500.4 ± 82.7	−39.5 ± 131.1	0.10	551.0 ± 106.1	514.5 ± 90.9	−36.5 ± 136.1	0.15	0.93
MDA (μmol/L)	3.7 ± 0.7	3.5 ± 0.6	−0.2 ± 0.5	0.007	3.9 ± 0.8	3.2 ± 0.6	−0.7 ± 0.9	<0.001	0.008

All values are means ± SDs.
ESR, erythrocyte sedimentation rate; FPG, fasting plasma glucose; GSH, total glutathione; HOMA-B, homeostasis model of assessment-estimated B cell function; $HbA1_C$, hemoglobin A1c; hs-CRP, high-sensitivity C-reactive protein; MDA, malondialdehyde; NO, nitric oxide; QUICKI, quantitative insulin sensitivity check index; TAC, total antioxidant capacity.
[a] P values represent paired-samples t-test.
[b] P values represent the time × group interaction (computed by analysis of the one-way repeated measures ANOVA).
[c] Obtained from McNemar test.
[d] Obtained from Chi-square test.

influence plasma NO and other biomarkers of oxidative stress. Previous observational and intervention studies have reported that supplemental vitamin D may have anti-inflammatory and antioxidant effect; nonetheless, results across all completed randomized trials have been inconsistent. Supporting with our study, supplementation of 60,000 IU/month vitamin D in rheumatoid arthritis patients with persisting disease activity and vitamin D deficiency for 3 months led to significant improvement in CRP levels (Chandrashekara & Patted, 2015). In addition, a 3-month treatment with 2000 IU/day vitamin D in Crohn's disease significantly decreased CRP levels (Raftery et al., 2015). However, our previous study among patients with major depressive disorder for 8 weeks have indicated that taking 50,000 IU/week vitamin D supplements did not influence hs-CRP levels, but increased TAC and GSH concentrations (Sepehrmanesh et al., 2016). Less production of parathyroid hormone following supplementation of vitamin D (Struglia et al., 2015) might result to a reduction in production of inflammatory factors including CRP. The accurate mechanisms by which vitamin D intake might influence biomarkers of oxidative stress are unknown. Improvement in the cellular GSH levels, decreasing the production of reactive oxygen species and pro-inflammatory markers by vitamin D supplements may explain the favorable effects on oxidative stress (42).

Table 4
Adjusted changes in metabolic profile of the patients with diabetic foot ulcer.

	Placebo group (n = 30)	Vitamin D group (n = 30)	P Value[a]
Vitamin D (ng/mL)	−0.005 ± 1.8	11.1 ± 1.8	<0.001
Ulcer length (cm)	−1.1 ± 0.2	−2.1 ± 0.2	0.001
Ulcer width (cm)	−1.1 ± 0.2	−1.9 ± 0.2	0.01
Ulcer depth (cm)	−0.5 ± 0.1	−1.0 ± 0.1	<0.001
FPG (mg/dL)	−9.3 ± 6.9	−35.5 ± 6.9	0.01
Insulin (μIU/mL)	2.0 ± 1.4	−2.6 ± 1.4	0.02
HOMA-IR	1.6 ± 0.8	−1.4 ± 0.8	0.009
HOMA-B	2.0 ± 3.1	−5.5 ± 3.1	0.09
QUICKI	−0.007 ± 0.003	0.007 ± 0.003	0.008
HbA1c (%)	−0.2 ± 0.1	−0.5 ± 0.1	0.04
Triglycerides (mg/dL)	−2.4 ± 5.7	−16.0 ± 5.7	0.09
VLDL-cholesterol (mg/dL)	−0.5 ± 1.1	−3.2 ± 1.1	0.09
Total cholesterol (mg/dL)	3.4 ± 4.7	−13.9 ± 4.7	0.01
LDL-cholesterol (mg/dL)	0.3 ± 4.4	−15.3 ± 4.4	0.01
HDL-cholesterol (mg/dL)	3.4 ± 0.9	4.9 ± 0.9	0.25
Total-/HDL-cholesterol ratio	−0.2 ± 0.2	−1.1 ± 0.2	0.003
hs-CRP (μg/mL)	2.2 ± 0.6	−0.7 ± 0.6	0.001
ESR (mm/h)	−19.1 ± 3.7	−33.6 ± 3.7	0.008
NO (μmol/L)	2.8 ± 1.0	3.3 ± 1.0	0.75
TAC (mmol/L)	−99.1 ± 32.1	−90.6 ± 32.1	0.85
GSH (μmol/L)	−44.7 ± 16.3	−31.3 ± 16.3	0.56
MDA (μmol/L)	−0.3 ± 0.1	−0.7 ± 0.1	0.01

All values are means ± SEs. Values are adjusted for baseline values, age and BMI at baseline.
ESR, erythrocyte sedimentation rate; FPG, fasting plasma glucose; GSH, total glutathione; HOMA-IR, homeostasis model of assessment-estimated insulin resistance; HOMA-B, homeostasis model of assessment-estimated B cell function; $HbA1_C$, hemoglobin A1c; hs-CRP, high-sensitivity C-reactive protein; MDA, malondialdehyde; NO, nitric oxide; QUICKI, quantitative insulin sensitivity check index; TAC, total antioxidant capacity.
[a] Obtained from ANCOVA.

The current study had few limitations. The sample size was small. Future studies with bigger sample size are needed to confirm the validity of our findings. Baseline HbA1c levels were different between the two groups. There was a significant improvement in HbA1c levels in the vitamin D group, which further complicated the situation such that it is would be difficult to extricate the effect of lowering glucose from that of vitamin D supplementation on wound healing, mainly as the baseline HbA1c levels were higher in the vitamin D group. It must be taken into account that there was a difference in serum vitamin D levels between the vitamin D and the placebo groups at study baseline. The difference in serum vitamin D levels between the two groups occurred at random. In addition, when we adjusted the analyses for baseline values of biochemical variables, no significant changes in our findings were observed. Although we agree that if they had similar levels to begin with, it would be even more interesting to know if supplementation of vitamin D made a difference to wound

healing. Future studies are needed to confirm our findings. Furthermore, in the current study, insulin and metformin therapy were not statistically significant different between the two groups. This data were taken from a 'yes/no' question. Since all patients were on insulin, it is very likely that the doses were changed during the study, which may have influenced results. Therefore, our findings should be interpreted with caution. Overall, vitamin D supplementation for 12 weeks among patients with DFU had beneficial effects on glucose homeostasis, total-, LDL-, total-/HDL-cholesterol, ESR, hs-CRP and MDA levels. In addition, vitamin D may have played an indirect role in wound healing due to its effect on improved glycemic control.

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